Code No: 135CC **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, November/December - 2018
ELECTRICAL ENGINEERING MATERIALS
(Common to ECE, CSE)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

Used paper 16-12-2018 PM

PART - A

(25 Marks)

1.a) Enumerate characteristics of dielectrics with respect to their valance band. [2]
b) What is meant by orientation polarization? [3]
c) Steel is ferromagnetic but is not a magnet in the absence of magnetic field why? [2]
d) What are the special purpose magnetic materials? [3]
e) What is meant by Bridgman method of semiconductor growth? [2]
f) Write the most fitting semiconducting materials for solar cells. [3]
g) Give the applications of brass and bronze as a low resistivity material? [2]
h) How do the lightly loaded contacts differ from heavily loaded contacts? [3]
i) What is meant by impregnation of the material? [2]
j) What is the effect of temperature on the insulating property of transformer oil? [3]

PART - B

(50 Marks)

2.a) What do you mean by dielectric loss? Explain loss factor, how do you calculate the loss factor.
b) Explain structure and properties of Barium Titanate as a promising ferroelectric material. [5+5]

OR

3.a) Explain how do you find out breakdown strength of solid dielectric material.
b) Briefly describe the significance of pyroelectric effect on pyroelectric devices. [5+5]

4.a) Explain the phenomenon of magnetic anisotropy and discuss its importance for magnetic materials.
b) Explain the concept of cast and cermet permanent magnets and use of such magnet. [5+5]

OR

5.a) Distinguish between soft magnetic material and hard magnetic material.
b) What are the different factors effecting permeability and hysteresis in magnetic materials? [5+5]

6. Discuss in detail, the stepwise methodology to manufacture the silicon wafer. [10]

OR

7. Write the most suitable semiconducting materials for integrated circuit and describe structure of LSI. [10]

8.a) What are the materials used for stranded conductors and write their composition.
b) Explain in brief characteristic features of thermocouple materials. [5+5]

OR

9. What are the properties of gaseous insulating materials and describe moisturizing effect on these materials? [10]

10.a) Describe the composition of radioactive materials and also its applications.
b) Describe the properties and applications of mineral oils. [5+5]

OR

11. With the help of suitable diagrams, describe stepwise, the method of galvanization of materials. [10]

---ooOoo---

Used Paper 16-12-2018PM